530.334 Heat Transfer – Spring 2026

Computer Project 2

Due date: Friday, April 10 at 10:00 am via GradeScope

BACKGROUND:

Solar panels are widely used to generate affordable and green energy. However, if they get heated for too long, the high temperature can cause fires in solar panels costing the equipment and risking lives. Thus, solar panels are equipped with cooling systems which passes ferrofluids through pipes inside a heat absorber layer to reduce the photovoltaic (PV) panel temperature. A simplified schematic of the PV and the coupled cooling system is shown in Figure 1 with glass layer on top and the heat absorber layer with the cooling tubes on the bottom of the PV panel.

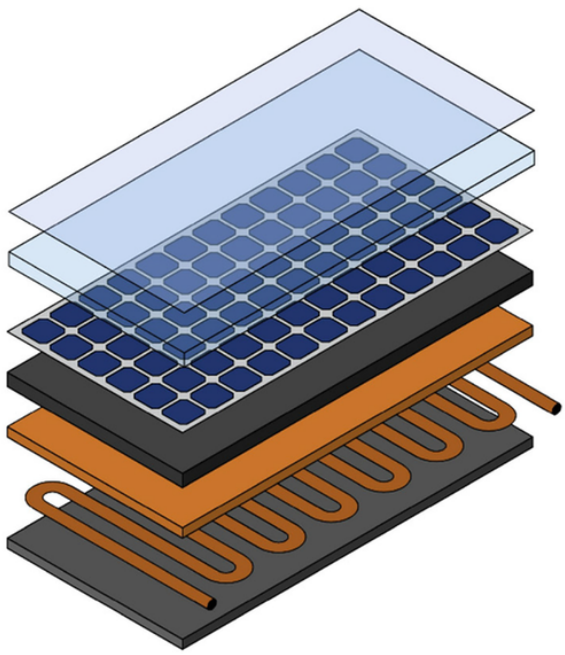

Figure 1 The simplified solar panel cooling system [1].

An engineer is assigned to the task to analyze the temperature distribution in a simplified cross-section of the solar panel cooling system as shown in Figure 2. Heat is absorbed at the top surface of the solar panel, while the side and bottom surfaces are insulated. Under steady-state operating conditions, the net radiation heat flux to the surface is $q''_{rad} = 900\ W\ /\ m^2$. Assume the length in the z direction is long enough that the temperature distribution is uniform through the z direction. The heat absorbing unit consists of five ferrofluid channels.

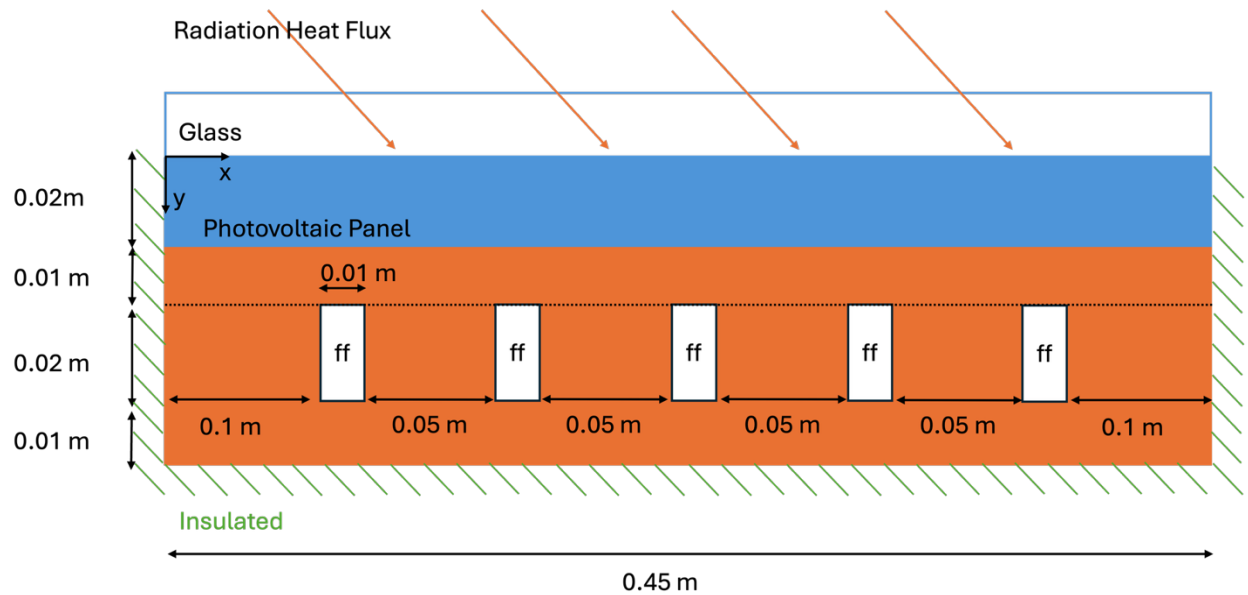

Figure 2 The cross section

DESCRIPTION:

- The thermal conductivity of the heat-conducting material (aluminum: orange) is $k = 205\ W/m \cdot K$ and the PV panel (blue) is $k = 10\ W/m \cdot K$.
- Thermal generation in PV panel is negligible when the solar heat flux is considered.
- The convection coefficient of the water is $h = 500\ W/m^2 \cdot K$.
- The width of the PV material is 0.45 m, and the height is 0.02 m.
- The width of the heat-conducting material is 0.45 m, and the height is 0.04 m.
- The width of each water channel is 0.01 m, and the height is 0.02 m. The position of the water channels is shown in Figure 2.
- Only radiative heat transfer occurs through the glass on top of the PV layer.

PROBLEMS:

1) Discretize the cross-section area of the heat absorbing material into 226 nodes in the x-direction and 31 in the y-direction. Use the energy balance method to derive the finite-difference equations for all types of nodes in this problem. Specify the boundary conditions.
2) Use MATLAB or Python to solve this problem and plot the temperature distribution across the cross section (x-y direction). Attach your code with proper comments. Find the highest temperature throughout this cross-section area and its location if the water temperature is 20 °C.

3) If the solar panel temperature should not exceed 130 °C to avoid the fire hazard, find the maximum temperature needed for the water temperature.

Required Materials for Submission: You are expected to turn in the governing equations, the plots as well as the code for all parts on Gradescope (no submission allowed through Canvas). Please submit any written work as well. The code file (.m or .py) should be submitted as is and compliable. Do not combine it with other materials or submit it as a PDF. Be sure that you include comments throughout your code.

Remember: All submitted materials should be your original work. Taking someone else's code (including your classmates, friends, Chat GPT, or other AI algorithms) and submitting it as if it is your work will be considered plagiarism and will NOT be treated lightly. If you have any questions about what does or does not constitute plagiarism, please ask the course instructor or teaching assistants.

References

[1] Bonilla, Amy & Hernández-Callejo, Luis & Redondo, Alberto & Martin, Karen & Alonso Gómez, Víctor & Aguilar, Carlos & Zorita, Angel & Sancho, Alejandro & Bassam, A.. (2025). Systematic Review of Flat Plate Photovoltaic Thermal Systems: Components, Efficiency, Monitoring, and Artificial Intelligence. Energy Technology. 13. 10.1002/ente.202500425.